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15 August 1984
SUPERSEDING
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MILITARY SPECIFICATION

TRUCK, FIREFIGHTING; AIRCRAFT AND PERSONNEL RESCUE, WATER AND FOAM DISPENSING, 1,000-GALLON WATER CAPACITY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of aircraft firefighting and personnel rescue truck with a capacity of 1,000 gallons of water and 130 gallons of foam agent.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

* 2.1.1 Specifications, standards, and handbooks. Unless otherwise specified (see 6.2), the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

O-D-1407 - Dry Chemical, Fire Extinguishing, Potassium Bicarbonate.
O-E-915 - Extinguisher, Fire, Dry-Chemical, (Hand-Portable).
TT-E-529 - Enamel, Alkyd, Semi gloss.

MILITARY

MIL-T-704 - Treatment and Painting of Material.
MIL-G-3787 - Glass, Laminated, Flat; (Except Aircraft).
MIL-F-24385 - Fire Extinguishing Agent, Aqueous Film-Forming Foam, (AFFF) Liquid Concentrate, for Fresh and Sea Water.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4210

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STANDARDS

FEDERAL

FED-STD-595 - Colors (Requirements for Individual Color Chips).

MILITARY

MIL-STD-281 - Automobiles, Trucks, Truck Tractors, Trailers and Trailer Dollies; Preservation and Packaging of.
MIL-STD-831 - Test Reports, Preparation of.
MIL-STD-1223 - Nontactical Wheeled Vehicles, Treatment, Painting, Rustproofing, Identification Marking and Data Plate Standards.

2.1.2 Other Government documents, drawings, and publications. The following other Government document form a part of this specification to the extent specified herein.

DEPARTMENT OF TRANSPORTATION (DoT)

Federal Motor Vehicle Safety Standards and Regulations.
Federal Motor Carrier Safety Regulations.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as Department of Defense adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN NATIONAL STANDARDS INSTITUTE, INC. (ANSI)

Z26.1 - Safety Code for Safety Glazing Materials for Glazing Motor Vehicles on Land Highways.

(Application for copies should be addresssed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

No. 412 - Evaluating Foam Fire Equipment Aircraft Rescue and Firefighting Vehicles.

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269.)

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SOCIETY OF AUTOMOTIVE ENGINEERS, INC. (SAE)

SAE Handbook.

(Application for copies should be addressed to the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

TIRE AND RIM ASSOCIATION, INC. (TRA)

TRA Yearbook.

(Application for copies should be addressed to the Tire and Rim Association, Inc., 3200 West Market Street, Akron, OH 44313.

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

* 2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. The truck shall be a cab forward, four wheel, four wheel drive, rear mounted diesel engine powered truck having a gross vehicle weight (gvw) rating of not less than 32,000 pounds (lb). The truck shall be equipped with a roof turret system, a bumper turret system, a handline system capable of dispensing AFFF or water, a Halon 1211 (liquified/gas) extinguishing agent handling system, a portable dry chemical system consisting of three 30-lb potassium dry chemical (P-K-P) extinguishers, a 1,000-gallon water tank, a 130-gallon foam tank, and related piping and other equipment and accessories as specified herein.

3.2 First article. When specified (see 6.2), the contractor shall furnish a truck for first article inspection and approval (see 4.2.1 and 6.4).

3.2.1 Configuration mock-up. A full scale configuration mock-up of the assembled vehicle cab, center body section, and firefighting system(s) shall be presented by the contractor for government inspection before final assembly of the first article. This configuration mock-up shall represent the design that will be used in the fabrication of the first article and shall demonstrate the interior seating, firefighting system(s), controls, and arrangement/human engineering relationship.

3.3 Standard truck, components, and accessories. Except as specified herein, the truck, components, and accessories shall be the manufacturer's standard or optional items which meet or exceed the requirements of this specification. All items shall be as represented and rated in the manufacturer's sales information, including special or mounted equipment. Sales information shall be limited to specifications and technical material identical to that furnished to authorized company representatives for selection of truck models and components.

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3.4 Interchangeability. All units of the same classification furnished with similar options under a specific contract shall be identical to the extent necessary to insure interchangeability of component parts, assemblies, accessories, and spare parts.

3.5 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification unless otherwise specified.

3.6 Performance. The fully equipped truck, with water, AFFF and Halon tanks filled to capacity, portable dry chemical extinguishers installed, a crew of four men in cab (weighing not less than 175 lb each or equivalent deadweight), and utilizing number two diesel fuel or JP-5 fuel, shall be capable of the following:

a. Ascending in both forward and reverse gears, a grade of not less than 30 percent while the transfer case is engaged in direct range and 60 percent while in low range.
b. Accelerating on a level road from a standing start to a speed of not less than 50 miles per hour (mph) in not more than 35 seconds.
c. On a level road, maintain a road speed of not less than 60 mph.
d. Ascending a 10 percent grade at a maintained 20 mph.
e. Maintaining 1.5 mph without vibration or engine stalling.
f. The noise level of the truck shall not exceed the values as specified in 3.6.2.
g. The truck shall develop 70 percent of maximum drawbar pull for 1 hour in low gear.
h. The carbon monoxide concentrations shall not exceed 100 parts per million in the cab during truck operation.
i. The truck shall have mobility characteristics for low speed operation over open fields, broken ground, loose sand, gravel, and general uneven cross-country terrain including drainage ditches. The truck shall traverse, in one pass, at low speed, terrain consisting of unflooded fine grained, dry to moist soils having a minimum cone index (CI) rating in the critical layer of no greater than 45 as specified in 6.5.
j. Starting and operating in ambient temperatures ranging from -25 degrees Fahrenheit (°F) to +125°F. Preheating accessories, such as glow plugs and ether starting aids, may be used for low temperature engine starting.
k. The service brakes shall control and hold the fully loaded truck on a 60 percent grade. Under all conditions of loading, service brakes shall stop the truck within 30 feet from a speed of 20 mph on dry,

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hard, reasonably level surfaces free from loose material. The service brakes shall be capable of fully recovering within 3 minutes at the end of mountain road test (3 miles of average 10 percent grades). A brake fade test shall show a deceleration rate of 15 feet per second at 30 mph to 0 mph at a stopping distance within 80 feet.

l. The handbrake shall hold the fully loaded truck on a dry concrete incline of 60 percent, and shall be capable of stopping the loaded truck from 30 mph at an average rate of deceleration of not less than 10 feet per second without demanding over 150 lb force applied to the actuation lever.

m. The truck shall be capable of being operated in both directions on side slopes up to 20 percent with extinguishing agents being discharged at maximum rated capacity from turrets without interruption; and remain upright when parked on a side slope of 30 percent, and the main engine shall be capable of being operated.

n. The truck shall satisfactorily negotiate an 18-inch vertical wall as described herein. Interference between truck components or between truck and obstacle profiles shall be cause for corrective action.

3.6.1 Altitude conditions. The truck shall satisfactorily deliver its rated pumping capacity at any elevation from sea level to 2,000 feet at an ambient temperature as specified in 3.6.

3.6.2 Sound level. The interior sound level shall conform to DoT Federal Motor Carrier Safety Regulations, section 393.94.

3.7 Functional characteristics. The truck shall incorporate such functional characteristics that will result in a practicable combination of the requirements specified herein with ease of production, safety, and comfort of personnel, accessibility for maintenance, and minimum requirements for special maintenance tools. All mechanical designs shall incorporate good human factors, engineering principles, and shall emphasize reliability, simplicity, rapidity, and straightforwardness of operation by personnel wearing protective clothing and under great stress.

3.7.1 Covers and access plates. Covers and access plates shall be provided where necessary, to facilitate inspection, servicing, repair, and adjustment. Covers and access plates shall be secured with the minimum number of fasteners compatible with sealing required. Fasteners shall be quick-release type wherever possible.

3.7.2 Protective features. In addition to accessibility, the installation of drain cocks for all air and fuel reservoirs shall be protected from accidental opening. The placement of all drains shall safeguard against drainage of contaminants on other components and accessories. Magnetized drain plugs shall be furnished for all gearboxes and oil reservoirs. Design of protective features shall:

a. Incorporate protection for fuel tank filler necks, door handles, brake cocks, flexible cables, pipe runs, and similar items to minimize damage resulting from use of these items for footsteps.

b. Assure that oil and fuel leakage or spillage does not contaminate other components, such as hose reels, batteries, auxiliary power

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unit, personnel heaters, wiring, and junction boxes during operation or servicing of the truck.

3.8 Weights, dimensions, and clearances.

3.8.1 Gross weight. The gvw rating shall be not less than 32,000 lb.

3.8.2 Weight distribution. With the truck fully equipped and loaded as specified in 3.6, the difference in axle loading shall not exceed 10 percent, and the difference in wheel loading shall not exceed 5 percent.

* 3.8.3 Dimensions and clearance. The truck fully equipped and loaded as specified in 3.6, on level ground, shall conform to the dimensions and clearances specified in table I.

TABLE I. Dimensions and clearances.

Overall length, inches (not more than)	336
Overall width, inches (not more than) 1/	108
Overall height, inches (not more than)	140
Ground clearance, inches (not less than) 2/	11
Turning clearance diameter, wall-to-wall, feet (not more than)	85
Angle of departure, degrees (not less than)	35
Angle of approach, degrees (not less than)	35
Wheelbase, inches (not more than)	195

NOTES: 1/ Overall width of the vehicle excluding outside rearview mirrors, shall be not more than 108 inches. The dimension from the outside of one wheel and tire to the outside of the opposite wheel and tire shall not exceed 112 inches.

2/ This dimension is acceptable providing vehicle off-road performance is not affected.

3.9 Chassis components.

3.9.1 Frame. The chassis frame shall be of bolted heat-treated steel construction and shall consist of longitudinal side rails with crossmembers, gussets located to provide adequate strength for the imposed loads, and maximum accessibility of all driveline components. Crossmembers shall be of either the rectangular or cylindrical, tubular type with box type weldment located at the truck frame to equalize stresses in all welds and in the crossmember assembly. The frame shall be designed to enable the truck to withstand all tests specified herein without permanent deformation. Composite basic frame structure, involving ferrous and nonferrous structural material, shall incorporate design features to minimize the effects of corrosion.

3.9.2 Tow hooks. Two heavy-duty, forged steel tow hooks shall be mounted at the front and rear of the truck. The hooks shall be attached to the chassis frame members or to rigid members which are attached to the frame.

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3.9.3 Engine. The truck engine shall be a diesel engine having the horsepower (hp), torque, and speed characteristics to satisfatorily meet all vehicular and pumping performance requirements specified herein. The engine shall be mounted at the rear of the chassis. The engine shall operate effectively on JP-5 fuel.

3.9.3.1 Air cleaner. The air intake for combustion shall be equipped with an oil bath or dry type air cleaner. The air cleaner, when tested in accordance with SAE J726, shall exhibit a cleaning efficiency of not less than 96 percent at a low engine air flow demand, with the efficiency increasing to not less than 97 percent in the 30 to 10 percent range of engine air flow demand. The cleaner shall exhibit no oil pullover characteristics at 100 percent engine air flow demand with the truck traversing cross-country terrain with 60 percent slopes. The air cleaner shall be accessible and easily removed for cleaning.

3.9.3.2 Oil filter. The engine manufacturer's standard full-flow oil filter of the replaceable cartridge type shall be provided.

3.9.3.3 Governor. A governor shall be provided that will limit the engine to the maximum full load revolutions per minute (rpm) recommended by the engine manufacturer. The governor shall provide maximum effectiveness under all engine operating conditions and shall not restrict engine power output below governed full load rpm.

3.9.4 Hour meter. An hour meter shall be provided to record hours of chassis engine operation. The hour meter shall be so mounted on the dash inside the cab, that it can be conveniently and easily read by personnel servicing the truck. Hour meter shall be of the oil pressure type.

3.9.5 Cooling system. The cooling system with circulating pump shall be of sufficient capacity to prevent coolant temperature exceeding 225°F in a pressurized closed system, as measured at the head outlet when operating continuously under normal operating conditions in an ambient temperature of 125°F. The radiator core shall consist of materials and construction in accordance with the best commercial practice, and shall be mounted so that it will not be damaged by vibration or frame distortion. An expansion tank of sufficient capacity shall be provided for the expansion of the coolant in the engine water jacket, the radiator, heat exchanger, and the heating system provided. Provisions for air separation shall be included. Thermostatically controlled, adjustable louvers shall be provided which will prevent the flow of air through the engine compartment when the engine is cold, and allow free air flow when the engine is warm. Shutters shall be fail safe in the open position if failure occurs.

3.9.6 Exhaust system. The exhaust system shall consist of an exhaust manifold and pipe(s), and a spark-arresting muffler or engine turbocharger. Means shall be provided for self-draining of the exhaust system when truck is parked in a level position. Location of the exhaust system shall eliminate the potential hazard of heat, carbon deposits, and fumes which are likely to adversely affect other components or operating personnel. Exhaust system shall be constructed of suitable corrosion-resistant materials.

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3.9.7 Fuel tank and lines. One fuel tank of not less than 50 gallons shall supply all fuel requirements. The tank and lines shall be amply protected against damage, excessive heat, and prevention of vapor lock. The tank shall be so mounted that it will not be damaged by frame distortion or collision with road obstructions, and shall not be effected by heat or flames that may surround the truck or from heat from the exhaust system. The fuel tank filler cap shall be the captive chained type. A compartment door shall provide easy access for insertion of filling nozzle. The fuel tank shall be equipped with a standpipe, 3/4-inch clearance from bottom of tank, for outlet fuel source to engine. Fuel tank shall have a drainage outlet located at the lowest part of bottom of the tank. Separate fuel pumps shall be provided for all components on the truck requiring diesel fuel. Fuel filters shall be included in the piping to each unit to prevent passage of foreign material.

3.9.8 Transmission and torque converter. A geared automatic or semiautomatic transmission, in conjunction with a torque converter, shall be provided. The transmission shall be as defined in SAE J645. The transmission shall provide for driver selection of individual gears or ranges of gears required for efficient operation throughout the speeds specified under the mobility requirement herein. The shift control shall incorporate a positive indicator for each gear or range selected. The transmission shall be so protected that the operator cannot inadvertently shift into reverse. The transmission assembly shall be of a commercial design and capacity that has been produced and supplied as part of a commercial or military vehicle, and shall have been field tested as a specific engine and transmission package. There shall be a minimum of four forward and one reverse speed ratios, properly spaced to provide the performance requirements specified.

3.9.8.1 Transmission capacity. The torque input capacity of the converter transmission assembly shall be in excess of the net torque output of the engine, and the maximum speed output of the engine shall not exceed the rated input speed of the transmission. The converter, with transmission design, shall acknowledge net flywheel hp.

3.9.8.2 Torque converter. The torque converter shall be heavy-duty having a multiplication factor of not less than two to one at stall. The torque converter shall be equipped with a heat exchanger. The torque converter transmission combination shall be provided with an oil filter.

3.9.8.3 Converter gage. A torque converter temperature gage, suitably mounted on the instrument panel, complete with a flashing warning light that will be clearly visible to the operator when the temperature of converter fluid exceeds the safe operating temperature, shall be provided. The signal light shall be suitably identified with a permanent metal plate. An additional audible device shall be provided which will actuate the horn when the converter fluid temperature exceeds manufacturer's recommended safe operating temperature.

3.9.9 Transfer case/differential. A two-speed transfer case/differential shall be provided. The differential shall be of the automatic or manual type. The manual type shall be remotely controlled from within the cab. All differentials shall be easily engaged.

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3.9.10 Universal joints. Universal joints shall be of the latest design and shall be identical make and type. They shall function satisfactorily under continuous operation at angles that will prevail with fully equipped and loaded
truck during all operational conditions.

3.9.10.1 Power train. All components utilized in the power train shall have torque capacity in excess of the maximum torque imposed by any gearing and the prescribed service conditions.

3.9.11 Axles. The axles shall be of the full floating type and of such a design and construction that in the event of an axle shaft failure at any point, the wheel will not be released. The rear axle shall be equipped with a manual or automatic, positive lock differential. The front axle assembly shall incorporate carbon type universal joints at each wheel for steering purposes. The front axle design shall preclude objectionable steering reaction of loud popping noise and jerking during tight turns. In order to achieve self-aligning and anti-shimmying effects, the steering axle shall be installed with positive caster and proper toe-in. The axles shall be designed for wide track, single tire mounting.

3.9.12 Suspension. The springs shall be heavy-duty, of adequate strength to sustain the maximum rated gross weight without evidence of overload or permanent set. Shock absorbers shall be furnished for the front and rear axles and have a minimum capacity as required for the imposed load. Auxiliary springs shall be provided, if required.

3.9.13 Power steering. Power assisted steering shall be provided and shall be such that a torque, not exceeding 15 lb applied at the steering wheel, will produce maximum efficiency of the power assisted steering at low engine speeds.

* 3.9.14 Wheels, rims, and tires. The truck shall be equipped with single front and rear wheels. Rim size and tire ratings shall conform to TRA recommendations. Disk-type wheels of the size suitable for the load and ground pressure shall be provided. Tires shall be of the wide base, highway type, and of rated capacity at least equal to the load imposed on each tire, measured at each wheel, at the ground, with the truck loaded to rated gvw and operating at maximum road speeds. Tread pattern of tires shall be super traction or road lug type. A spare wheel with mounted tire shall be furnished, but shall not be mounted on the truck.

3.9.15 Braking system. The truck shall be equipped with four-wheel, full air application or air hydraulic-type brakes with quick buildup system. The braking system shall be capable of safely controlling the fully equipped and loaded truck under all operating conditions. The system shall be pedal operated from the driver's position and conveniently located so that the driver, wearing rubber boots, can safely activate the brake pedal without interference from other control pedals or cab obstructions. Ease of accessibility for servicing and adjustment of brakes is essential. The design shall include adjustment features without removal of the wheels, and shall include shields to safeguard against accumulation of mud and water in the brake drum. The brake system shall be independent from the accessory system, firefighting system controls, transfer case differential lock, in order to eliminate the possibility of a brake failure in the event of a malfunction in

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the accessory system. Power from the generator shall be provided to the brake system on a first priority over the accessory system requirements. The braking system, complete with all necessary components, shall include:

a. A 12 cubic feet per minute minimum rated compressor with an oil bath or dry air cleaner on the intake side.
b. Air storage reservoir(s) with not less than 2,400 cubic inch capacity, each tank equipped with drain and with safety and check valves between compressor and last reservoir tank.
c. Foot control, relay, and safety valves.
d. Air pressure gage visible to the driver.
e. Low pressure warning indicator, visible and audible to the driver.
f. An air dryer of at least four stage process to effectively remove water, oil carbon, and sludge from the air brake system.

3.9.15.1 Air brake emergency system. When the vehicle is equipped with full air application brakes, an emergency brake system shall be furnished. The system shall be capable of being manually applied by the operator seated in the driver's seat. A manual or mechanical means to release the brakes shall be provided when compressed air is not available to release the brakes. The emergency system shall also serve as a parking brake and shall conform to DoT Federal Motor Carrier Safety Regulations, section 393.41.

3.9.15.2 Split hydraulic brake system. If the vehicle is equipped with air hydraulic brakes, it shall be equipped with a service brake system so arranged as to provide separate systems for at least two wheels, so designed and constructed that rupture or leakage type failures of the brake master cylinder body, effectiveness indicator body, or other housing common to the divided system will not result in complete loss of function of the vehicle brakes when force on the brake pedal is continued. Pressure component means any internal component of the brake master cylinder or master control unit, wheel brake cylinder, brake line, brake hose, or equivalent.

3.9.15.3 Parking brake. When the vehicle is equipped with air hydraulic brakes, an independent mechanically operated parking brake mounted on the drive shaft shall be provided, capable of meeting the performance requirements as specified in 3.6.

3.9.16 Electrical system. The electrical system shall be 12 volts (V) nominal direct current (dc) potential. Wiring shall be thoroughly secured in place, shall be insulated, splash proofed, and protected against exposure to ground fires. The electrical system shall include an alternator-rectifier system, batteries, switches, circuit breakers, lighting, a rotating beacon, a siren-speaker system, and horn.

3.9.16.1 Alternator. An alternator-rectifier system shall be furnished complete with voltage regulator. The alternator shall be rated for not less than 145 ampere (A).

3.9.16.2 Battery(s). Battery(s) furnished shall be 12V potential, of the sealed cell (maintenance free) type, and shall be rated in terms of cold cranking A capacity at 0°F in accordance with SAE J537. The battery system

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shall have a total cold cranking A capacity of not less than the engine manufacturer's recommendation of cold cranking A required at 0°F for the engine furnished. The battery(s) shall be easily accessible for servicing and removal. The battery cables shall be protected by use of grommets where the cables pass through metal bulkheads.

3.9.16.2.1 Battery charger/auxiliary power system. The battery charger/auxiliary power supply unit shall contain all components necessary to convert 115V, alternating current (ac), 25A electrical power to 12V, dc 50A power, and to maintain a required level of charge to the batteries from an external 115V, ac power source available from the junction or terminal box on the truck. The battery charger shall be of an automatic control type that will monitor the proper battery cell voltage and provide the necessary charging rate at all times. When battery comes up to full charge voltage, the unit shall provide a low trickle charge of 0.25A rather than shut off. Overload and short circuit protection shall be incorporated to protect the unit during engine starting. The battery charging unit shall be incorporated into a waterproof and salt water corrosion-resistant box mounted near or in the truck's battery compartment. A means shall be provided to receive external 115V ac power through a receptacle mounted at the rear of the truck. The receptacle shall be a three pole, male cast housing, insulated terminal back arrangement with spring loaded sealing cover, similar or equal to Joy Manufacturing X8042-8.

3.9.16.3 Circuit breakers. All electrical circuits shall be protected by reset circuit breakers of the automatic trip type to provide for overload current protection. Circuit breakers shall be readily accessible and protected against excessive heat, physical damage, and water spray.

3.9.16.4 Lighting. The truck shall be equipped with clearance lights, parking lights, stoplights and taillights, and directional signals. All lights shall be of sufficient size and shall be in accordance with DoT Federal Motor Carrier Safety Regulations. In addition, the lighting shall conform to the following:

a. Two sealed beam headlamps. Headlamps shall be mounted as low as practicable, beams shall be adjustable and shall be adjusted so that center of upper beam will strike the ground 300 feet ahead of fully loaded truck on level ground.
b. Directional signal lights shall be of the self-canceling type with visual and audible indicator. The controller shall be mounted on the steering column and shall be wired through the ignition system. A switch for 4-way emergency flashing shall be provided.
c. Cab dome lights with independent switches shall be provided.
d. All gages and valve controls shall be properly lighted and readily visible from the operator's position.
e. A red rotating beacon, equipped with not less than two sealed beam lamps, shall be installed on the top front left corner area of the truck so as to provide maximum visibility over 360°. The control switch shall be installed on the cab instrument panel and be independent of the dash switch.
f. All lenses on front mounted headlights shall be glass 1/.

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g. A 500-watt light, similar or equal to Wide Light (Model L-500-DM), shall be installed on the front centerline of the truck above the bumper turret. The control switch shall be installed on the cab instrument panel. All necessary associated electrical equipment shall be provided for proper operation of this light.

NOTE: 1/ The lenses on the front mounted lights (turn signals, reflectors, and marker lights) may be other than glass providing they have the highest temperature melting characteristics available.

3.9.16.5 Siren and speaker system. A combination electronic siren and speaker system shall be mounted on the top front right corner area of the truck. The unit to be Penetrator Model PT75 or equal, and shall be controlled within the cab and within reach of the driver.

3.9.16.6 Horn. An electrical horn actuated by a button in the hub of steering wheel shall be provided.

3.9.16.7 Wiring. All wiring shall be encased in conduit, loom, or wiring harness, and shall be color coded or number identified. The harness or individual wiring shall be protected by the use of grommets where they pass through metal bulkheads.

3.9.17 Instruments and controls.

3.9.17.1 Driver's instrument panel. Instruments and controls shall be grouped by function insofar as practicable and shall be located so that they can be seen from the driver's position. All instruments and contols shall be suitably identified by metal plates with letters at least 1/4-inch high in contrasting colors. All controls shall be capable of effective operation by the 5th through 95th percentile of personnel wearing crash clothing and rubber boots. The main instrument panel shall contain the following:

a. Speedometer and recording odometer.
b. Tachometer for foam pump engine.
c. Ammeter or voltmeter for vehicle driven engine.
d. Fuel gage.
e. Coolant temperature indicator for vehicle driven engine.
f. Oil pressure gage for vehicle drive engine and transmission.
g. Oil temperature gage for transmission.
h. Ignition switch (no key required).
i. Ignition and starting switch for drive engine.
j. Ignition and starting switch for pump engine.
k. Master switch for accessories.
l. Light switches except floodlights, compartments, and trouble lights.
m. Headlight beam indicator.
n. Start/stop switch for pump engine.
o. Windshield wiper and washer controls.
p. Low air pressure signal.
q. Air pressure gage, green and red sector.
r. Pump engine (oil pressure gage, coolant temperature gage).
s. Pump and drive engine hour meters.

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t. All instruments shall be lighted indirectly and controlled by a separate switch so that the instruments can be lighted without turning other lights on.
u. All gages and instruments shall be marked with red and green lines or sections as appropriate to indicate normal operating conditions or "do not exceed values."

3.9.17.2 Equipment operator's control panel. An equipment operator's control panel shall be provided in the cab at the right side (passenger side) for housing all control instruments, gages, and instruction plates applicable to firefighting system performance as controlled by the roof and bumper turret operator and other related systems. Commercially available gages which incorporate more than one function are acceptable. This control panel shall include but not be limited to the following:

a. Duplex (combination) water and foam pump pressure gage (red lined in hazardous operating area).
b. Foam tank level gage or tank level indicator lights indicating full, 3/4, 1/2, 1/4, and empty (electronic type 0-130 gallons).
c. Water tank level gage or tank level indicator lights indicating full, 3/4, 1/2, 1/4, and empty (electronic type 0-1,000 gallons).
d. Booster heater controls.
e. Other pertinent controls and instruments.

3.9.18 Cab.

3.9.18.1 Cab construction. The cab shall be an all aluminum structure of welded construction, enclosed forward single compartment with ample room for a four-man crew. Interior of body sides, roof, and doors shall be treated with an insulating material and shall be lined with washable, high strength, pressed wood fiber, marlite, plastic, or metal panels. The access steps to cab interior and exterior shall be so designed as to provide maximum safety for personnel wearing crash clothing when entering or leaving the cab. The cab shall be rainproof and dripproof, and shall provide for maximum visibility. The cab shall be tested to a water spray simulating a rainfall of 3 inches per hour for period of 1 hour to determine leakage.

3.9.18.1.1 Cab doors. The cab shall have one door on each side; all cab doors shall be designed so that they can be opened and closed from inside and outside. Wide-opening, sliding type doors shall be provided with the necessary steps and handgrabs to permit rapid, safe entrance and exit from the cab. Automatic locking of the doors in the fully opened and closed position with manual release shall be provided. All doors shall have design features which will prevent the doors from becoming damaged under any condition of vehicle operation required herein. The outside door handles shall be not more than 6 feet above ground level and shall not protrude appreciably beyond the body skin.

3.9.18.1.2 Cab seating. A seat shall be provided for the driver and shall be a four-way adjustable seat not less than 20 inches wide. Individual seating spaces for two crewmen shall be provided in the cab; one seat shall be located at the right rear facing forward; and one seat (with fold-down platform) shall be located facing forward adjacent to the driver. A third crewman folding seat

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located behind the driver's seat shall be provided. Individual seats shall be padded seats of suitable width and depth to provide ample seating width for crewmen in turnout coats. A turret operator's seat shall be provided adjacent to the driver's seat facing forward and shall be used by the firefighting equipment operator for both roof and bumper turret operation. The back of this seat shall fold down to provide a platform structure for the turret operator while standing in the roof hatch opening. All seats shall be upholstered over polyurethane foam cushion with first grade Naugahyde, or equal, suitable for fire service. Seat belts shall be furnished and installed for the driver and for each seated position provided. Seat belt length shall be made to accommodate crewmen wearing turnout coats. Storage space shall be provided under crewmen seats.

3.9.18.2 Windows. The cab windshield shall be laminated, safety glass conforming to MIL-G-3787, class II. The width of the windshield shall be the maximum consistent with the truck width. Windows of tempered, safety glass per ANSI Z26.1 shall be provided in the cab and doors. All windows shall be capable of sliding open or rolling down.

3.9.18.2.1 Visibility. The design and arrangement of the cab and components contained therein shall be such as to afford optimum visibility for control of the truck and performance of firefighting operations. The operators shall be able to see the ground at a point not greater than 20 feet from the truck, through an included angle of at least 105° on each side of the centerline of the windshield (210° total angle). The restriction of the horizontal angle of vision by obstructions, such as window frames, corner, and door posts shall not exceed 7° each. Forward vision through the windshield between the forward corner posts of the cab shall be unobstructed, except the center post between the operators is permissible if the center post width is restricted to that width (in all cases under 2-1/4 inches) which imposes no blind area at a distance of 40 feet from the truck when viewed by either the driver or equipment operator sitting in a fixed, normal operation position. Upward visibility at least 15° above the horizontal is required at eye level.

3.9.18.3 Windshield washers and wipers. The windshield wipers shall be heavy-duty dual wipers of the electrical type with capacity and coverage to readily cleanse the windshield of rain, snow, ice, and foam when used in conjunction with defrosting system or windshield washer mechanism. Wiper arms and blades shall be of sufficient length to clean at least 70 percent of the windshield area. Wiper assemblies shall be synchronous and shall include features to permit speed control and return of the wiper blades to a retracted position out of the line of vision when not in use. Windshield washer system shall produce a high volume blast of spray for cleaning foam in the path of the wiper blades. The washer reservoir shall have a capacity of at least 1 gallon and the container must be nonbreakable. All controls shall be in reach of the driver from his normal driving position.

3.9.18.4 Cab roof. The cab roof shall contain one access hatch conveniently situated to permit free turret nozzle operation by the turret operator standing in the cab on a platform. The height of the platform shall be such as to enable the turret operator to be positioned in the roof hatch opening so as to grasp the turret nozzle handle at shoulder height (approximately 58 inches from platform deck to turret handle for average turret operator). The opening shall be of such a size to permit personnel,

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while wearing heavy protective clothing, to open the top latches quickly and man the turret that is to be installed on the vertical centerline of the roof. The opening shall be free of sharp objects and adequately padded to provide protection to personnel who may be standing on the turret operator's platform. The opening shall incorporate a side hinged upswing door or a sliding door that is equipped with a quick release opening mechanism and shall include an automatic restraining device, which will hold the door in the fully open position and prevent damage to the door when slammed into the open position. The fully opened hatch door shall be positioned to the right of the turret operator when the turret operator is positioned in hatch opening. Suitable handholds or a grab handle shall be provided near the hatch opening and secured into the cab roof to enable the turret operator to maintain his balance when standing in the open hatch while the vehicle is maneuvering. A suitable grab handle(s) shall be provided and secured on the inside of the hatch cover for ease in closing. The cab roof shall be designed and constructed to provide sufficient strength and rigidity to sustain the loads imposed by the mounted equipment and the personnel required to service the equipment.

3.9.18.5 Heating system. A coolant type heater shall be provided in the heating system. Heater shall operate satisfactorily on diesel fuel supplied from the truck engine fuel tank. The liquid distribution system shall be adequate to protect personnel, agents, and equipment during exposures to ambient temperatures of -25°F. Heat output for the cab shall be sufficient to provide and maintain a temperature of +40°F at the driver and crew's feet position at an ambient temperature of -25°F after 1 hour of heater operation. Adequate heat to maintain a temperature of +40° shall also be supplied to the remainder of the truck to protect those agent tanks, lines, and equipment components subject to freezing during standby operation at -25°F. Fluid lines shall not be drained during these operations. Cab heating system shall meet the requirements of DoT Federal Motor Carrier Safety Regulations, section 393.77, as applicable.

3.9.18.5.1 Power plant heater. An engine coolant heater shall be provided. The heater shall operate on 110V, ac, 60 Hertz (Hz), and shall be wired through a junction block to a single three pronged (male) weatherproof slave receptacle for receiving external power and grounding vehicle. The receptacle shall be located on the pump operator control panel. A three-wire connecting cable, 25-feet long, and of adequate capacity to supply power to the heater, shall be furnished. The connecting cable shall include a matching female connector at the vehicle end, and a standard three-pronged (two power plus one ground) male connector at the other end. Electrical apparatus shall conform to DoT Federal Motor Carrier Safety Regulations, section 393.77(c), (7). The electrical insulation of the connecting cable shall withstand normal operating stresses in low ambient temperatures (down to -65°F) without cracking or loss of dielectric capacity. All heater lead wires shall be installed without interfering with vehicle component operation and without loose excess wire. Provisions for stowage of the cable shall be provided in the vehicle cab. The heater furnished shall be as follows:

a. A coolant heater shall be installed in the engine block or lower coolant hose. A coolant circulating pump driven by a 110V, ac, 60 Hz motor shall be provided. The coolant heater shall be a 2,250 watt, 19.5A, 115V, 60 Hz, ac, heater, Kim "hot-start", Model BC-110 or equal.

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3.9.18.5.2 Defrosting, anti-icing, and ventilating. A sufficient defrosting, anti-icing, and ventilating system shall be provided in the cab. This system shall have the capacity to effectively prevent formation of frost in the interior surface of all transparent areas of the cab and to prevent the exterior formation of ice on the windshield of the vehicle. The system shall also maintain clear vision over at least 75 percent of the area of all windows when the cab interior is at a mean temperature of 40°F and the ambient temperature is -25°F. Fresh air intakes, louvered from the outside, shall be provided as a means of diverting the output air to a point outside the cab in order for infinite heat input control and prevent overheating with moderate outside temperature.

3.9.18.5.3 Controls. The heat supplied to body area, protecting agent tanks, lines, and equipment components, shall be thermostatically controlled from 40°F to 75°F. Variable speed controls shall be provided on air blowers and at least high and low output settings shall be furnished on cab heating system controls. Adequate operating instructions shall be provided at heater system control point. Cab heater and defroster system control shall be mounted on the instrument panel.

3.9.18.6 Cab air conditioning system. The truck cab shall be cooled by a standard automotive-type air conditioning system. The air conditioner shall be designed for use with dichlorodifluoromethane (refrigerant 12). The air conditioner shall be sized to reduce the cab temperature to 85°F within 30 minutes and maintain the temperature while the truck is in a ready standby condition with engine at idle speed and exposed to 115°F ambient temperature. Flexible transitions shall be provided between the air conditioner evaporator supply and return air openings and the truck cab. The air conditioner shall be capable of operating in the normal cooling mode (not on bypass) at all engine speeds within the range of ambient conditions described on a psychrometric chart by the following limits:

a. The 40°F dry bulb line to intersect with its line of saturation.
b. The saturation line from 40°F to 84°F.
c. A line from 84°F saturation point to 97°F dry bulb and 87°F wet bulb.
d. A line from 97°F dry bulb and 87°F dry bulb to 125°F dry bulb and 75° wet bulb.
e. The 125°F dry bulb line to 125°F dry bulb and 75°F wet bulb.

3.9.18.6.1 Air conditioning compressor. An automotive type compressor powered by the truck engine shall be provided. The compressor shall be of sufficient capacity to ensure compliance with the cooling requirements specified herein.

3.9.18.6.2 Air conditioning condenser. The condenser shall be an air-cooled type with tube and fin construction. The condenser shall withstand 1-1/2 times the saturated refrigerant pressure at 160°F without rupture or distortion.

3.9.18.6.3 Air conditioning evaporator. The evaporator shall be of tube and fin construction and shall withstand an internal pressure of 1-1/2 times the saturated refrigerant pressure at 160°F without rupture or distortion. The evaporator shall have adequate surface and tube distribution to ensure the cab temperature requirements specified in 3.9.18.6.

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3.9.18.6.3.1 Evaporator fan. The evaporator fan shall be sized and selected to ensure compliance with the cooling and noise requirements specified herein. The fan shall be driven by a multispeed motor that obtains power from the vehicle electrical system.

3.9.18.6.3.2 Evaporator drain. The evaporator shall be equipped with a drip pan and drain for overboard disposal of condensate water.

3.9.18.6.4 Controls. The air conditioning system shall incorporate safety controls for protection of the system in accordance with SAE J639, and shall have adequate controls for starting the air conditioner and controlling air temperature from within the truck cab.

3.9.18.6.5 Piping. Refrigerant piping shall be sized to provide minimum pressure drops throughout the system. Maximum practical use shall be made of flexible refrigerant lines and quick disconnect couplings between major components.

3.9.18.6.6 Noise level. When measured as specified herein, the noise level generated by the air conditioner system shall not exceed the following values:

Octave band center Frequencies (Hz)	Octave band pressure level (In decible ref. 0.0002 microbar)
63	85
125	80
250	76
500	73
1,000	71
2,000	70
4,000	69
8,000	68

3.9.18.7 Rearview mirrors. An external combination (flat and convex) mirror, enclosed in a common housing and each with at least four supporting arms or heavy-duty, spring loaded, ratchet type mounting brackets, shall be mounted on each side of the cab. Each combination mirror shall have at least 50 square inches of flat reflective area and an adjustable convex surface having at least 20 square inches of reflective area. Each mirror shall be viewable from the operator's position.

3.10 Body. The body shall consist of two sections; a center section which shall enclose firefighting system(s) and equipment with the exception of the turrets (which are cab mounted) with related piping and controls, and a rear section which shall consist of engine enclosures, two sets of fixed steps located at the rear, leading to the top of the truck with top and side panels. The body shall be made of aluminum alloy having not less than 42,000 pound-force per square inch (psi) ultimate and 35,000 psi yield strength and a thickness of not less than 0.050 inch. The center body section shall be insulated and heated to assure that the compartments (see 3.10.2) temperature will not fall below 40°F with an ambient still air temperature of -25°F. All body insulation shall be enclosed. Body lines shall be flush with no projecting valve handles, knobs, and pipes. The fixed steps of the rear body

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section shall have minimum width of 12 inches, 12 inches deep, with a maximum height at the rear bottom step of 25 inches and constructed of sufficient strength and rigidity to sustain shock loads imposed by a 200 lb fire fighter jumping on the steps. The steps shall be constructed of nonskid type material. Handholds and rails shall be provided for safe personnel movement on and off the top of the body. Handrails shall be secured so that the rails will not rotate within bracket attachments. The right and left sides of the truck body shall be extended to a height of at least 12 inches above the working deck and shall be fitted with chromium-plated or stainless steel handrails. The center body section shall be of the unitized one piece type. The center body section shall be three point rubber mounted to the truck frame. Compartments shall be integral one-piece type and shall be three point rubber mounted to the truck frame. The center body section and the hose bed design shall permit easy removal and replacement of the firefighting engine, pumps, and clutches from the right side and/or the top of the truck.

3.10.1 Working deck. The section behind the cab area shall be a combination working and storage space equipped with a door for maintenance of the firefighting system module. It shall be of reduced height and capable of carrying light cargo of 50 pounds per square foot loading. The deck shall be nonskid type material and attached to the center body section in such a way as to be easily removable.

3.10.2 Compartments. Suitable compartments within the center body section shall be provided for the convenient storage and removal of equipment and tools furnished. Compartments for the storage of items which need exposure protection shall be weathertight, insulated, and heated. Doors shall be reinforced and of slam-shut design. Operating handles shall be of chromium-plated steel, operable with heavy-gloved hands. Top hinges on doors on emergency equipment compartments shall have spring operated props for one-hand operation. Drip rails shall be installed above all compartment doors. All compartments shall be provided with adequate lighting and the lighting shall be controlled by door switches. Provisions shall be incorporated to permit flushing and cleaning inside of compartments.

3.10.2.1 Stowage equipment. An enclosed compartment shall be provided for stowage of accessory equipment. The size of this compartment shall be such that three 30 lb dry chemical extinguishers as specified in 3.11.4 could be stored. Location of this compartment shall be opposite driver's side of truck.

3.11 Firefighting system and equipment.

3.11.1 General. The main firefighting system shall consist of water and foam concentrate tanks, foam/water dispensing turret nozzles, water and foam pumps with drive engine, agent proportioning system with appropriate piping, and system controls to meet the firefighting performance requirements specified herein. A secondary system consisting of a foam/water producing handline system and a Halon 1211 liquid/gas agent system shall also be provided. In addition, portable extinguishers with dry chemical and other miscellaneous equipment as specified herein shall be provided.

3.11.1.1 Firefighting system performance. The vehicles firefighting system, when in pumping mode, shall discharge foam and/or water from storage tanks within the vehicle body through the roof and bumper mounted turrets and

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the handline system. The vehicle shall be able to discharge agent while maneuvering around the fire site. The firefighting system shall provide for selective operation of roof turret, bumper turret, and handline nozzle for properly proportioned foam or water. Agent flow through the roof turret shall be 500 gallons per minute (gpm) (±50 gpm); the bumper turret shall have a flow rate of 250 gpm (±25 gpm); and the handline shall have a minimum flow rate of 60 gpm (+15 -0 gpm) at 100 psi nozzle pressure (+25 -0 psi).

3.11.2 Agent (foam/water) application and foam proportioning system.

3.11.2.1 Fire pump. The unit shall incorporate a 1,000 gpm single-stage centrifugal fire pump for agent (foam/water) application. The suction inlet of the pump shall be lower than the bottom of the water tank to assure gravity flow to the pump under all conditions specified herein. The pump shall be removable as an assembly without disturbing other components. Mechanical pump seals, if provided, shall be so constructed and mounted that water will not enter into any gearbox or transmission in the event of pump seal failure. The pump shaft shall be fabricated of corrosion-resistant steel and the impeller shall be bronze. The pump casing shall be cast iron or aluminum. For test purposes, the fire pump shall be capable of independent manual operation without engagement of the foam pump.

3.11.2.2 Foam concentrate pump. The pump shall be either a single-stage centrifugal or positive displacement type driven by the auxiliary pump engine. The suction inlet of the pump shall be lower than the bottom of the foam tank to assure gravity flow to the pump under all conditions specified herein. The foam concentrate pump drive shall be so arranged that the water pump must be operating any time the foam concentrate pump is operating. For test purposes, the foam pump shall be capable of independent manual operation without engagement of the fire pump.

3.11.2.3 Pump drive system. The fire (foam/water), 1,000 gpm pump and the 250 gpm foam concentrate pump shall be driven by an independent diesel engine through a remotely controlled hydraulically, pneumatically, or electrically operated clutch. Engagement of the engine throttle and firefighting system valves (opening and closing) shall be controllable from the cab by the driver in the normal seated position. This same capability shall be provided at the manual override control station as specified in 3.11.2.4. The fire and foam pumps shall be directly driven by the pump engine. Belts and pulleys are not acceptable. The hp, torque, and speed characteristics of the diesel engine shall be compatible with the pump design characteristics. An hour meter shall be provided for the pump engine and shall be easily accessible for all engine maintenance operations. The starting motor for the pump engine must be suitable for operating in a 12V electrical system. Removal of clutch assembly from auxiliary pump engine shall be permitted without removal of the pumping unit.

3.11.2.4 Pump system controls. The control system for the pumps, except for engine starting, shall be pneumatically or electrically operated. All components shall be capable of being operated manually in case of loss of pressure. Manual controls to valves must be easily accessible with adequate performance clearances, free from encumbrance, and so located to permit

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personnel to reach and manually operate control levers during emergency situations. Manual controls must be identified at the valves with "on-off" operating procedures clearly marked. A separate 2,400 cubic inch reservoir, isolated through a back pressure regulator, shall supply the control system and prevent bleeding down of the vehicle brake air supply. On-off controls shall be provided for each turret operator and the roof turret controls shall be located immediately adjacent to the hatch opening. After pump engine has started, each pump control system shall cause proper sequential operation of clutch engagement, valves, engine throttle, and all other components to provide smooth start-up and shut-down operations without hydraulic or mechanical shock to the system and without discharge of liquids within the vehicle. The maximum allowable time lag, from throwing the control to the foam position until full turret output, shall be 4 seconds; likewise, the maximum allowable overrun time, from throwing the control to the off position until no turret discharge, shall be 4 seconds. All manual override controls shall be located at the pump panel area and shall be accessible for one man operation at one station position.

3.11.2.5 Pump inlet piping. Inlet piping for the water and foam pumps shall provide for continuous flow of liquid to the pumps without cavitation at all flow rates up to rated system capacity. Tank shutoff valves shall be provided in the inlet piping of both the water and foam pumps. The foam and water tank shutoff valves shall be controlled at the left side of the truck. The foam tank shutoff valve shall be located to minimize wetted piping. The foam tank inlet piping shall include a check valve to prevent backflow under any condition and shall be positioned near main shutoff valve.

3.11.2.6 Agent discharge piping. The discharge piping shall incorporate the water pump pressure relief bypass system, the pumper discharge connections, and the foam concentrate pump bypass system. Both the water and foam discharge piping shall prevent entrance of water into the foam concentrate piping and components, and prevent foam entrance into the water piping and components upstream of the proportioning system under all conditions specified herein.

3.11.2.7 Water pump pressure relief bypass. The pressure relief bypass shall incorporate a pump to water tank return line, pump bypass, and control valves. The pressure relief bypass shall ensure that proper system pressure is maintained for operation of the turrets and handline nozzles at rated capacity in any on-off combination within 3 seconds after any nozzle is turned on or off with any engine throttle setting above that required for maximum agent system performance.

3.11.2.8 Pump discharge connections. Two 2-1/2-inch national hose (NH) male discharge connections shall be located on the left side of the truck and shall terminate in a side compartment near the engine/pump enclosure. Each connection shall incorporate a quarter turn valve with nonsticking ball and self-adjusting seats. The valve shall be self-locking in any position. Each connection shall be provided with a chained-on brass or chromium-plated brass rocker lug cap. The connections shall be no more than 48 inches above the ground. Piping from the fire pump to the connections shall be sized corresponding to the size of the connections without the use of reducers.

3.11.2.9 Pump suction connections. The water pump inlet piping shall include two 2-1/2-inch NH suction valves. The valves shall be quarter turn, nonsticking ball with self-adjusting double seats to hold pressure or vacuum

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in either direction. The suction valves will have female swivels with removable strainers, self-locking handles, and chained-on brass or chromium-plated brass rocker lug caps. The pump suction connections shall be located in the side compartment below the engine/pump enclosure on the driver's side of the truck. The connections shall be used for pumping from a resupply vehicle or from a hydrant. Piping to the fire pump from the connections shall be sized corresponding to the size of the connections without the use of reducers.

3.11.2.10 Foam proportioning system. On the pump discharge side, an automatic balanced pressure proportioning system, to control the ratio of foam concentrate to the quantity of water being discharged shall consist of:

a. A pump to deliver the foam concentrate at a slightly higher pressure than that of the water on the discharge side of pump.
b. A pressure control valve to automatically bypass sufficient foam concentrate back to the storage tank to produce and maintain a foam concentrate pressure exactly equal to the water pump discharge pressure.
c. Calibrated metering valves in the foam concentrate lines shall be designed to maintain any percentage from 0 to 6 percent. Meters shall be factory set at 6 percent.
d. Proportioning eductors installed in such a manner to ensure good mixing of foam concentrate and water and obtain accurate proportioning.

3.11.2.11 Foam quality. The foam quality of AFFF shall meet the requirements of MIL-F-24385, and shall have a minimum expansion of three with a drainage time of not less than 1 minute when sampled in accordance with NFPA No. 412.

* 3.11.2.12 Agent tank(s). The agent tank shall be referred to as a fiberglass reinforced, polyester foam/water tank for crash truck operation. The tank shall consist of 1,000-gallon water chamber with an integral or separate rigid and removable 130-gallon foam cell. The water tank should be molded and configured so the bottom reinforcing structure will provide a locking barrier to prevent lateral shifting of the tank from the subframe. The tank bottom shall be designed to provide 80 percent delivery of contents of both water and foam cells on a 20 percent grade. The water tank shall be baffled longitudinally as well as transversely at distances not greater than 36 inches. Similarly, the foam cell shall be baffled horizontally to allow static loading of the water chamber before foam filling. The foam cell shall be a totally enclosed, self-supporting unit. The foam cell and its contents must be thermally protected by a minimum of two inches of water between the front wall of the water tank and the foam cell. Outlets shall be so designed to provide complete drainage of the foam cell and water tank. Tank outlet fixtures shall have molded bases with threaded stainless steel inserts to allow pipe flange attachments. Inserts shall be molded into the bases in such a way to allow full tightening torque without coming loose or cracking. No thermoplastic couplings or fittings shall be used through either tank wall. Water tank baffles shall be bolted through tank sides, top, and bottom. Side bolting shall be through 3/16-inch mild steel, flat on tank, outside through tank, and baffle and stainless steel flat inside. Bolting shall be 3/8-inch

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stainless steel on 6-inch centers, sides, and top. The water tank shall be fitted with four steel hold-down brackets attached in such a way as to provide bolting through the longitudinal baffles, hold-downs to be of sufficient strength to retain the tank through spring loaded assemblies to the subframe. The foam/water tank shall be provided with lifting lugs to enable the tank to be removed from the truck. The foam/water tank shall be fitted with a water fill box capable of receiving two 2-1/2-inch line flange connections. The cover shall be bolted and include a water level viewing window. The foam cell shall contain a quick removable filler funnel and ten mesh screens capable of delivery of poured foam concentrate to within 12 inches of foam cell bottom. Foam cell shall have a hinged quick-lock cover. Water and foam cells shall be fitted with provisions for overflow capacity to match inlet flows. Similarly, both units shall have provisions for level central gages and temperature probes. Minimum entry to water and foam tanks shall be 15 by 15 inches. Both tanks shall be tested statically. The foam cell and water tank shall withstand the full foam/water load at 5 psi and at all conditions of loading for one hour with minimum wicking. The laminate resin system shall be polyester resin with 42 percent ±1 percent stryene for the water tank and chemical bonded chorlean corrosion-resistant resin for the foam cell. Minimum for loading shall be 28 percent. Maximum laminate shall be 1/4-inch thickness. The foam/water tank shall be mounted on a subframe (torque box) that is turn mounted to the truck frame. The subframe shall be constructed of aluminum. It shall contain a rubber-lined longitudinal surface that fits into a corresponding area recessed into the tank, thereby eliminating lateral movement of the tank. The tank shall be spring mounted on all four corners of the subframe in order to absorb acceleration and braking shock loads.

3.11.2.12.1 Agent tank vent and overflow system. The water tank shall incorporate an overflow system that will allow sufficient overflow when filling water tank from two 2-1/2-inch intakes. The foam tank shall incorporate at least a 2-inch diameter pipe overflow system. The overflow systems will relieve excess liquid when the tanks are being filled. Each tank shall incorporate a vent system which is combined within the overflow system. The vent systems provide for relieving positive and negative pressure on the tanks during tank fill and agent system discharge. The overflow and vent system prevents leakage of water and foam liquid from the tanks when filled to capacity and operating on the maximum side slopes and grades and mobility conditions specified herein and during all agent system pumping operations. The vent and overflow systems shall not be affected by any of the environmental conditions specified herein including freezing conditions.

3.11.2.12.2 Agent tank drains. The agent tanks shall incorporate drain pipes and shutoff valves. The drain valves shall be controlled by an operator at the left side of the truck. The points of discharge of all drains shall be below the chassis frame. The foam tank drain shall be located so the contents of tank can be drained into 5-gallon cans. The water tank drain shall be at least 3 inches in diameter. The foam tank drain shall be at least 1-inch in diameter.

3.11.2.12.3 Water tank access box. A water tank access box 15 by 15 inches shall be provided in the top of the tank and shall contain a bolted cover with a clear transparent sight opening of at least 6 inches in diameter.

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3.11.2.12.4 Water tank loading connection. The tank shall incorporate two 2-1/2-inch water intake connections at the rear of the truck. The connections shall be provided with 2-1/2-inch quarter turn, gated valves with female NH hose connections. These female connections shall be enclosed with waterway screens and rocker lug plugs chained to the valve. The fill line shall be mounted to preclude accumulation of water and freezing at low temperatures. The connections shall be no higher than 48 inches from the ground, readily accessible, and boldly marked to denote location. These two fill lines are to fill the tank through the access box.

3.11.2.12.5 Foam tank filler box. The foam tank shall incorporate a filler box mounted in the top of the tank readily accessible for at least two operators on top of the truck. The filler box shall be at least 6 inches high and incorporate a hinged cover and latch to prevent spillage under any operating condition. The filler box shall be designed to be filled by an exterior transfer hose of 1-1/2-inch diameter at any flow rate up to 75 gpm. The tank shall incorporate an overflow provision to indicate when it is filled to capacity. The filler box shall be capable of allowing two standard 5-gallon foam liquid containers to be emptied simultaneously. The filler box filler neck shall extend near enough to the bottom of the tank to reduce foaming to a minimum during the filling operation. The filler box shall incorporate a removable strainer to prevent entry of foreign matter into the tank during all filling operations. The strainer shall be rigidly constructed and at least 10 mesh.

3.11.2.12.6 Foam tank loading system. Foam tank loading shall be accomplished from the existing 250 gpm foam pump. An electrical priming pump and accessory filling hose shall be provided. The fill line shall be mounted to preclude accumulation of foam and freezing at low temperatures. The filler line shall fill the tank via a line which terminates in an antifoam funnel-shaped baffle below the foam concentrate surface. The foam tank filler inlet shall be located in the agent system control and stowage compartment as described in 3.11.3.

3.11.2.12.7 Tank level gages. Tank level gages shall be provided for both the water and foam concentrate tanks. The gages shall be mounted inside the cab in a readily visible location to the driver. The gages shall be of the electronic type. The gages shall allow a determination of quantity of agent in the tanks with an accuracy of $\pm$5 percent of the individual tank capacities at any level from empty to full. Tank level indicator lights which indicate tank fill condition of full, 3/4, 1/2, 1/4, and empty may be furnished in lieu of gages.

3.11.2.13 Agent discharge turrets. A roof and bumper turret shall be installed at the front of the cab to provide for frontal approach to the fire. All nozzles and piping thereto, from the valves out, shall be vented and automatically drained when the valves controlling discharge to the nozzles are in the off position to provide fast draining to prevent freezing during low temperature operation. Means shall be provided to prevent the turrets from inadvertently leaving their storage positions due to truck motion, yet not restrict instantaneous employment when necessary.

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3.11.2.14 Roof turret. The roof turret shall be installed forward on the cab roof centerline and shall be positioned to allow for the turret to be operated manually by the turret operator standing in the hatch opening in the cab roof. The roof turret shall have a nonaspirating constant flow, variable stream nozzle capable of discharging either AFFF agent or water selectively at a dual flow rate of 500/750 gpm (±50 gpm). The quality of foam (AFFF) produced shall be as specified herein. The nozzle shall provide for straight stream or dispersed pattern. The minimum reach of the straight stream shall be 150 feet from nozzle tip.

3.11.2.14.1 Roof turret control. The roof turret discharge valve shall be a manually actuated, foam/water selection, on-off remote control valve located within easy access on the left of the turret operator standing in the hatch opening. Also, external controls for infinitely variable pattern, from straight stream to dispersed, shall be provided to the turret operator, mounted on or part of the turret nozzle directional control handle. Control handles shall be of sufficient size and shape to allow turret operator to grasp with heavy gloves. Stream pattern control shall be hydraulic, pneumatic, or mechanical. If a hydraulic or pneumatic control system is used, a means shall be provided to vary nozzle pattern upon loss of power assist. The horizontal arc of straight stream agent coverage shall be at least 210° symmetrical about the centerline of turret rotation. The vertical arc of straight stream agent coverage shall extend from the position where the nozzle discharge just clears the truck to an elevation of at least 60° above the horizontal. The turret shall depress to a position to be capable of providing specified foam coverage on an arc beginning at least 30 feet in front of the truck and mounted at a height so that there shall be no nozzle obstruction during right or left swing.

3.11.2.15 Bumper turret. The bumper turret shall be a remote hydraulically operated turret mounted on a small platform in the front of the cab on the vehicle centerline. Control of the bumper turret shall be accomplished by a hydraulic/electric valve control arrangement that will allow three-way hydraulic or hydraulic/electric valve control inside the cab within the reach of the driver. The nozzle of the bumper turret shall be a nonaspirating, constant flow type with a fully variable pattern from straight stream to fully dispersed. The nozzle shall be capable of dispensing AFFF agent or water. The AFFF shall be of the quality as specified herein. The rated discharge capacity of the turret shall be 250 gpm (±25 gpm). The minimum reach of the straight stream discharge shall be 150 feet from the nozzle tip.

3.11.2.15.1 Bumper turret nozzle control. The bumper turret (monitor type) nozzle shall be remotely controlled hydraulically/electrically from within the cab. The power assist system shall be independent from any other hydraulic system installed on the truck. A servo-mechanism shall be provided permitting full control of the turret nozzle for direction of the stream in both elevation and rotation. The control handle in the cab shall be positioned to operate in phase with the turret for both rotation and elevation. Pattern control shall be accomplished with a toggle switch arrangement. A single push button control shall be used for on-off flow of agent. The lever control and on-off control valve shall be accessible for operation of the truck driver in his normal operating position. Horizontal control shall allow the vertical arc of straight stream agent coverage to include an arc from at least 45° above horizontal to a depressed position capable of providing a foam agent coverage

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on an arc beginning at least 20 feet in front of the truck, through the full horizontal movement of the turret. The pattern control shall allow an infinitely variable pattern from straight stream to dispersed.

3.11.2.16 Handline system. A combination of foam/water agent handline system shall be provided consisting of a hose, a hose nozzle, a hose reel, and hose reel controls. The handline nozzle shall be of the constant flow, variable stream type, and shall have a solution discharge capacity of 60 gpm at 100 psi and capable of producing foam at the quantity specified herein, and be capable of dispensing the agent in an infinite variation of patterns between a solid straight stream and full spray. The nozzle shall have a heavy-duty bail type handle with dual stops, nonstick ball shut off, and a flush without shutting down capability. The straight stream shall have an effective reach of at least 60 feet. The handline shall consist of 150 feet of 1-inch, 800 psi test rubber-lined hose, made up in one complete section with NH threaded male and female couplings.

3.11.2.16.1 Handline hose reel. The handline hose reel shall be capable of containing 150 feet of 1-inch hose as specified herein and shall be located in a compartment opposite the driver's side, positioned so as to allow handline hose withdrawal to the front of the truck. The hose reel shall be designed for electric rewind with a manual means for rewinding as a backup. A rewind control switch shall be positioned to allow one person to activate the switch and feed the hose on the reel. The switch shall be protected from water entrance. The design of the hose reel assembly shall be such as to prevent dragging, snarling, and overrunning of the reel during unreeling of the hose and agent discharge operations. During hose reel unwind, the hose reel electric motor shall not be engaged. A manually adjustable friction brake to prevent snarling and overrunning of the reel shall be provided. The braking system shall be disengaged manually before activation of the rewind motor. A means shall be provided for restraining the nozzles in a readily accessible stowed position that will allow easy access by a fire fighter while standing on the ground. The hose reel shall be of a durable, rigid construction as light in weight and compact as possible. An air purge system, with check valves to prevent water from entering the truck air system, shall be provided to purge the hose to prevent freezing under low temperature conditions. The air purge system control shall be located in the handline nozzle compartment.

* 3.11.2.17 Halon agent system. A Halon 1211 liquified gas type agent system shall be provided. The system shall consist of an agent tank with a capacity of 500 lb of agent; a tank of high pressure nitrogen for expelling the agent and purging the system, and electric rewind hose reel assembly fitted with 150 feet of 1-inch, 800 psi test rubber-lined hose, made up in one complete section with NH threaded male and female couplings, and a special Halon nozzle and related control valves, gages, and piping. The electric rewind hose reel shall be mounted in a side body compartment. The Halon nozzle shall be two position type, vapor discharge and straight stream. The nozzle shall discharge at a rate of 5 - 5-1/2 lb per second and shall deliver a straight stream with a range of 35 feet and a vapor stream range of 10-15 feet from the end of the nozzle.

3.11.2.18 Firefighting system control. The truck firefighting system control shall include all manual and automatic controls required for operation of the agent system as specified herein. Automatic controls shall be

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pneumatically, hydraulically, or electrically operated. The controls shall be arranged to allow full operation of the agent system for crash firefighting by the equipment operator in a normal seated position. Driver access to controls shall be as specified herein. The agent system shall exhibit no instability in pressure or flow rate under any operating condition.

3.11.2.18.1 Pressure surge. The control system shall incorporate surge arresting devices to prevent pressure surges under any condition which could cause damage to the truck after numerous occurences. The control system shall be capable of allowing simultaneous shutoff of the turret nozzles and the handline without exceeding the normal governed speed of the engine.

3.11.2.18.2 Control system maintenance adjustment. The control system shall be so designed and installed that all control adjustments can be accomplished in a specified sequence and all adjustments are independent. The control system shall be so designed that all adjustment points are accessible without the removal of any parts or components.

3.11.2.19 Drain system. In addition to the agent tank drains specified herein, the agent system including all components shall incorporate drains to provide for rapid and complete draining of the system. Once drained, the truck shall withstand temperatures down to -25°F without causing any damage to the agent system or any of its parts or components. Careful attention shall be given to drainage gage and control line tubing to prevent freeze damage. Drainage shall not be allowed to flow over or onto other truck components. Rubber hose at least the size of the drain valve shall be provided to conduct the flow to the outside of the truck enclosure. Drain valves which are not readily accessible from beside the truck shall be provided with remote controls which will permit draining without crawling under the truck. Drain valves shall be grouped to reduce the number of locations. Drain lines shall be grouped to reduce the number of valves. In addition to the agent tank drains, no more than six drain control locations shall be required to completely drain the system. The drains shall be sized so the entire agent, other than the agent tanks, can be drained in no more than 4 minutes.

3.11.3 Agent system control. An enclosed compartment shall be provided directly behind the vehicle cab on the driver's side below the pump engine compartment. In addition to the agent system discharge outlets and pump inlet specified in 3.11.2.12.6, this compartment shall house the controls and instruments necessary for pumping mode procedures, hydrant and hose operations, water and foam tank filling.

3.11.3.1 Control panel. The control panel shall be provided in the compartment within easy reach and clearly visible to an operator standing on the ground. The control panel shall be easily accessible from the rear for maintenance. All instruments and controls shall be illuminated with nonglare lighting to be readily visible for night operation. The control panel shall include at least the following instruments and controls:

a. Fire pump pressure gage (30 inches mercury (hg) vacuum to 400 psi).
b. Foam pump pressure gage (30 inches hg vacuum to 400 psi).
c. Water and foam pump relief valve controls.
d. Pump engine tachometer (at least 3-1/2 inches in diameter).

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e. Pump engine throttle (pull type with push button latch and turn for microadjustment).
f. Pump engine oil pressure.
g. Pump engine coolant temperature.
h. Panel light controls.
i. Water tank temperature gage.
j. Foam tank temperature gage.
k. Pressure gages (0-400 psi) for #1 and #2 discharge outlets.

3.11.3.2 Pressure gages. Pressure gages shall be at least 4-1/2 inches in diameter and flush mounted type. Each gage shall be provided with a throttle valve and control located directly beneath the gage to allow elimination of pressure pulsations. The gages shall have face-mounted adjustment and removable front glass. The gages shall have no less than 20 psi graduations. The gages shall have a 2-1/2 percent accuracy of the full scale range with the vacuum and pressure ranges considered individually. Each gage line shall incorporate a 1/4-inch national taper pipe thread plugged test connection on the control panel.

3.11.3.3 Transverse hose storage bed. Two transverse hose beds shall be located above the pump engine compartment area and each capable of storing 200 feet of 1-1/2-inch double-jacketed, cotton, rubber-lined fire hose. The transverse hose storage bed shall be an all enclosed compartment with a top door for loading and side doors for removal of the hose to either side of the truck. Hose payout or removal opening shall be provided with hose rollers installed in the bottom and both sides of each transverse bed opening to prevent hose interference or snagging. The bottom of the transverse hose beds shall be slotted, well seasoned oak or other hardwood having equal characteristics. Discharge outlet controls shall be located in the agent system control compartment described in 3.11.2.8 and 3.11.3. The two 1-1/2-inch hose beds shall be provided with the necessary foam/water supply piping to allow a flow of 95 gpm to each hose bed. A 1-1/2-inch NH swivel hose connection shall be provided in the center, bottom of the hose beds. Adequate space for hose hook-up and swivel movement shall also be provided. Each hose bed connection shall have a special pressure gage in the pump panel. Gages and valves shall be identified as 1-1/2-inch (#1) and 1-1/2-inch (#2).

3.11.4 Portable dry chemical system. Three 30-lb capacity dry chemical fire extinguishers in accordance with O-E-915 (size 30, type II, class 2), shall be furnished. All extinguishers shall be mounted in an unused storage compartment opposite driver's side of the truck. The fire extinguishers shall be charged with P-K-P conforming to O-D-1407. All extinguishers shall be mounted with suitable carrying brackets in the compartment so as to prevent shifting during truck travel. All three extinguishers shall be readily removable and operable.

3.11.5 Piping and valves. Piping valves shall be metal that is fully compatible with the AFFF concentrate, water, and their admixtures. Reinforced rubber hose may be used in lieu of piping for fluid transfer except where rigidity is required for mounting of valves and other inline devices. The hose shall be compatible with foam concentrate and solutions. Bolted type, split housing couplings or unions of the victaulic type shall be used in making

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connections, to facilitate removal and replacement. If conventional flanges are necessary, all bolts shall be easily accessible. All piping and reinforced hose shall be properly braced and supported. All piping, couplings, and valves shall be sized for required flow with minimum restriction and friction loss. Piping shall be securely mounted and provided with flexible couplings to minimize stress.

3.11.6 Winterization drains. In addition to the tank drains, all piping and components subject to freeze damage when the truck is stored in cold weather shall be provided with adequate drain points. Drain valves shall be operable from the sides of the truck or accessible through openings in the pump housing located in the truck cab, and the control of each valve shall be identified in 1/2-inch high contrasting letters.

3.11.7 Corrosion preventatives. The construction and connections shall be made to prevent the possibility of galvanic corrosion of dissimilar metals. When the use (connection) of dissimilar metal cannot be avoided, they shall be separated by suitable insulating material. Materials subject to the corrosive and weathering effects of salt or other atmospheric conditions and the extinguishing agent shall be suitably protected. Parts normally exposed to contact with foaming agents in concentrate form shall not be made of light metal alloys.

3.12 Auxiliary firefighting equipment. The auxiliary equipment shall be stored in compartments providing the required temperature and weather protection. The necessary compartments shall be provided with necessary holders and brackets to secure the equipment firmly in place under all operating conditions, and yet permit rapid and easy removal on arrival at the site of the emergency. An adjustable wrench, hydrant and spanner wrench, a forcible entry claw tool, Halligan or equal, shall be furnished. A ladder mounted on the right hand side of the vehicle with a quick-release bracket shall be provided. The ladder shall be of the combination "A" style step ladder, aluminum, 7-1/2 feet long folded, or 14-1/2 feet long locked open, similar or equal to Duo-Safety ladder model 35A.

3.13 Treatment and painting. Vehicle surfaces shall be cleaned, treated, and painted in accordance with MIL-T-704, type A. The surfaces to be painted shall be primed with zinc chromate epoxy applied to a dry film thickness of 1.5-2.5 mils. The primer used shall be compatible with the final paint system. The finish paint shall be polyurethane enamel or equal, applied to a dry film thickness of 1.8-2.2 mils. The finish color shall be yellow gloss No. 13538 in accordance with FED-STD-595, and shall be applied to the exterior surfaces of the truck cab, chassis, body, and to interior surfaces of the compartments. The cab interior shall be primed and finish painted with two coats of semigloss enamel conforming to TT-E-529. The finish color shall be green No. 24525 in accordance with FED-STD-595. The disk wheels shall be painted the same as the exterior color. Rims and tire locking rings treated with permaplate or other similar corrosion-resistant treatment need not be painted. Radiator grille, headlight rings, and other exterior trim shall match the exterior color specified, except that chromium-plated, cadmium-plated, stainless steel, or anodized aluminum surfaces need not be painted.

3.13.1 Rustproofing. The vehicle shall be rustproofed in accordance with MIL-STD-1223.

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3.13.2 Finishes and protective coatings. Metal parts shall be conditioned, primed, coated, and finished in accordance with the best practices of the fire apparatus industry. Materials used shall be of first class quality and designed to give long lasting protection under the intended service conditions.

3.14 Lubrication. Unless otherwise specified (see 6.2), means for lubrication shall be in accordance with the manufacturer's standard practice. The lubricating points shall be easily visible and accessible. Hydraulic lubrication fittings shall be in accordance with SAE J534. Where use of high-pressure lubricating equipment, 1,000 psi or higher, will damage grease seals or other parts, a suitable warning shall be affixed to the equipment in a conspicuous location. The unit shall be lubricated prior to delivery with type of lubricant specified in the operator's manual and grade of lubricant recommended for ambient temperature at the delivery point. The unit shall be conspicuously tagged to identify the lubricants and their temperature range.

3.15 Servicing and adjusting. Prior to acceptance of the truck by the Government, the contractor shall service and adjust the truck for immediate operational use as required in the operator's manual. The servicing and adjusting shall include at least the following:

1. Inflation of all tires.
2. Adjustment of brakes.
3. Proper functioning of all lighting and electrical systems.
4. Wheel alinement.
5. Adjustment of engine to include tune-up.
6. Complete lubrication with grades of lubricants recommended for ambient temperature at the delivery point.
7. Cooling system filled to capacity with a clean solution of equal parts by volume of water and antifreeze (ethylene glycol).

The truck shall be conspicuously tagged to identify the lubricants and their temperature range.

3.16 Electromagnetic interference characteristics. The truck chassis shall conform to the electromagnetic characteristic requirements and test limits of SAE J551.

3.17 Marking and data plates.

3.17.1 Identification plate. The contracting officer will furnish to the Government inspector the required identification plates. The contractor will be required to stamp the necessary data in the blank spaces thereon and securely affix said plates in a conspicuous place on each unit, assembly or subassembly, and parts as directed by the Government inspector. Nonferrous screws, rivets, or bolts of not less than 1/8-inch in diameter shall be used to affix the plates. Nomenclature shall be "TRUCK, AIRCRAFT FIREFIGHTING AND RESCUE."

3.17.2 Equipment marking. All emergency used items and compartments shall be plainly identified with bold, contrasting black block leters 1-inch in height. All nonemergency used items, compartments, access hatches, and drain points shall be identified with block letters 1/2-inch in height. Tank

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capacities shall be indicated on hatch covers. The levers and controls within the cab shall be clearly indicated as to function by means of engraved metal plates resistant to obliteration by wear. Instruction and identifying plates shall be located so as to be visible to the operator's from their normal positions.

3.18 Workmanship.

3.18.1 Steel fabrication. The steel used in fabrication shall be free from kinks, sharp bends, and other conditions which would be deleterious to the finished product. Manufacturing processes shall not reduce the strength of the steel to a value less than intended by the design. Manufacturing processes shall be done neatly and accurately. All bends shall be made by controlled means to insure uniformity of size and shape.

3.18.2 Bolted connections. Boltholes shall be accurately punched or drilled and shall have the burrs removed. Washers or lockwashers shall be provided in accordance with good commercial practice, and all bolts, nuts, and screws shall be tight.

3.18.3 Riveted connections. Rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet heads, when not countersunk or flattened, shall be of approved shape and of uniform size for the same diameter of rivet. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member.

3.18.4 Welding. Welding procedures shall be in accordance with a nationally recognized welding code. The surface of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the parts connected by the welds. Welds shall transmit stress without permanent deformation or failure when the parts connected by the weld are subjected to proof and service loadings.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.1.1 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements specified herein and in applicable referenced documents.

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* 4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1).
b. Quality conformance inspection (see 4.2.2).

* 4.2.1 First article inspection. The first article inspection shall be performed on the truck when a first article is required (see 3.2 and 6.2). This inspection shall include the examination of 4.3 and the tests of 4.4 through 4.8.7. The first article may be either a first production item or a standard production item from the supplier's current inventory provided the item meets the requirements of the specification and is representative of the design, construction, and manufacturing technique applicable to the remaining items to be furnished under the contract.

* 4.2.2 Quality conformance inspection. The quality conformance inspection shall include the examination of 4.3, the tests of 4.10, and the packaging inspection of 4.11.

4.3 Examination. Each truck shall be examined for compliance with the requirements specified in section 3 of this specification. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirements or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

4.4 Tests.

4.4.1 Preparation for test. The first article truck submitted for test shall contain all specified components and shall be completely assembled and properly prepared for operation. The truck, including all components, shall be correctly adjusted, lubricated, and serviced, and all equipment which has been ordered shall be mounted for immediate operation as specified in 3.15. Fuel and agent tanks shall be full unless otherwise specified for a particular test.

4.4.2 Data. During the tests specified herein, at least the following data, unless not applicable, shall be recorded at intervals not greater than 15 minutes or as otherwise indicated. Additional data may be required and recording intervals decreased for test of short duration or as otherwise dictated by test objectives.

a. Date.
b. Time started.
c. Time finished.
d. Ambient temperature.
e. Atmospheric pressure.
f. Agent level in the water tank.
g. Water tank temperature.
h. Agent level in foam concentrate tank.

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i. Foam concentrate tank temperature.
j. Water pump inlet pressure (in Hg).
k. Water pump discharge pressure.
m. Agent flow rate through each nozzle.
n. Engine speed (rpm).
o. Engine oil temperature.
p. Engine water temperature.
q. Torque converter oil temperature.

4.4.3 Test observations. Throughout all tests specified herein, the truck shall be closely observed for the following conditions which shall be considered cause for rejection.

a. Failure to conform to performance requirements specified herein.
b. Spillage or leakage of any liquids including firefighting agents, coolants, and lubricants under any operating condition.
c. Structural failure of any part including permanent distortion or evidence of impending failure.
d. Evidence of excessive wear.
e. Interference between truck components or between the truck and the ground and all required obstacles with the exception of normal contact by the tires.
f. Misalignment of parts and components.
g. Instability in agent system pressure of flow.
h. Evidence of undesirable roadability features including instability in handling during turning, braking, and when traversing rough terrain.
i. Conditions which present a safety hazard to personnel during operation, service, or maintenance.
j. Overheating of the truck engine or any components.

4.5 First article test report. Upon completion of the first article tests, a test report shall be prepared in accordance with MIL-STD-831.

4.5.1 Reliability and maintainability information. The following information shall be included as an appendix to the test report:

a. All failures, servicing, adjustments, maintenance, and irregular functioning shall be identified by accumulated operating time, cycles, miles, or position in the test procedure as appropriate. Test conditions at the time of the events identified shall be recorded.
b. A summary of the engineering analysis and of any test conducted to determine assignable causes for any failure or irregular functioning.
c. A summary of the engineering analysis leading to any corrections made to design, construction, quality control, or other procedures or leading to any corrections to be made or proposed to be made to production items. The summary shall also include an analysis of the predicted effectiveness of such corrections.

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d. Clock time and man-hours required for each maintenance and servicing action taken during the tests.

4.6 First article firefighting system tests.

4.6.1 Features and components. All features and components shall be inspected for proper installation, adjustment, and serviceability. The truck shall be driven for a distance of at least 5 miles over paved highway and at an average speed of 40 mph and the agent system, including all nozzles individually and in all combinations, shall be operated while the truck is in motion. All components shall be moved through their maximum normal displacement during applicable operations of driving and agent system discharge, and shall exhibit no malfunctioning, leaks, objectional rough operation, or irregular movement.

4.6.2 Agent tank(s).

4.6.2.1 Agent tank(s) static tests. The foam cell and water tank shall be filled with foam and water, then subjected to an internal static pressure of 5 psi for not less than one hour. Tests shall also be performed with the water tank full and foam tank empty; and water tank empty, foam tank full. A soap solution shall be sprayed on the exterior of the tanks to facilitate detection of leaks that may exist. Evidence of leakage shall be cause for rejection.

4.6.2.2 Water tank filling. The empty water tank shall be filled from a 2-1/2-inch hose placed in the filler box at an average flow rate of at least 250 gpm. The filler box shall be checked for proper size strainer. The quantity of water required to fill the tank shall be measured. At the end of the filling cycle the tank overflow device shall be checked.

4.6.2.2.1 Water tank rear loading. The empty water tank shall be filled with water from two 1-1/2-inch hoses at the back of the truck at an average flow rate of at least 250 gpm each.

4.6.2.2.2 Water tank side filling inlet (hydrant). It shall be demonstrated that the tank can be filled from pump suction connections located in driver's side compartment specified in 3.11.2.9.

4.6.2.3 Foam concentrate tank filling. The empty foam concentrate tank shall be filled from the 250 gpm foam concentrate pump at an average flow rate of at least 75 gpm. The quantity of foam concentrate required to fill the tank shall be measured. The filler box shall be checked for a required strainer, and the overflow device shall be observed in use.

4.6.2.3.1 Foam tank loading internal system. It shall be demonstrated that the foam tank can be loaded through the on board pump and piping system described in 3.11.2.12.6.

4.6.3 Agent system foam discharge. The truck shall be operated to demonstrate that the agent system will produce foam from the roof turret at a flow rate of at least 500/750 gpm; the bumper turret at a flow rate of at least 250 gpm; the handline at a flow rate of at least 60 gpm; the 2-1/2-inch discharge outlets at a flow rate of 250 gpm; and the two 1-1/2-inch outlets at

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a flow rate of 95 gpm, each using foam concentrate in accordance with MIL-F-24385. The tests shall be conducted for each nozzle individually in a straight stream pattern and in the widest dispersed pattern. The foam discharge from each nozzle shall be analyzed to verify that the foam concentrate is between five and seven percent of the total volume of agent. The foam discharge shall be analyzed to determine foam quality of at least three expansion and having a 25 percent drainage time above 1 minute when measured as specified in NFPA No. 412. The agent system shall be observed to determine that there is no instability in pressure or flow rate under any operating condition. Each flow condition shall be maintained for a minimum of 30 seconds.

4.6.4 Agent system water discharge.. The truck shall be operated to demonstrate that the agent system will discharge water from the roof turret at a flow rate of at least 500/750 gpm; the bumper turret at a flow rate of at least 250 gpm; the handline at a flow rate of at least 60 gpm; the two 2-1/2-inch discharge outlets at least a flow rate of 250 gpm each; and the two 1-1/2-inch outlets at a flow rate of 95 gpm. The tests shall be conducted for each nozzle individually in a straight stream, widest disperse pattern, and for straight stream of nozzle in roof, bumper and handline combination. Included cone angles of at least 45° shall be measured for turrets in the maximum dispersed pattern position. The agent system shall be observed to determine that there is no instability in pressure or flow rate under any operating condition. Each flow condition shall be maintained for a minimum of 3 minutes.

4.6.5 Water tank exhaustion. The capability to discharge 97 percent of the water tank capacity by normal means when the truck is on level ground and 80 percent when the truck is headed in any direction on a 20 percent slope shall be demonstrated.

4.6.6 Roof turret test. The roof turret shall be tested to determine compliance with 3.11.2.14 and 3.11.2.14.1. The turret system controls shall be tested for performance and ease of operation. With the aid of appropriate measuring devices, the range of turret movement shall meet the specified requirements. The freedom of movement, convenience of operating height, and accessibility of controls shall be tested.

4.6.7 Bumper turret test. The bumper turret shall be tested to determine compliance with 3.11.2.15 and 3.11.2.15.1. The range of turret movements shall be measured to determine compliance with specified requirements.

4.6.8 Handline tests. The handline nozzle shall be checked for proper stowage. Water discharge rates, concentration and foam, and pattern characteristics shall be met as specified herein. During operations, the controls shall be checked for adequacy and ease of operation. The handline shall be run fully out under simulated emergency conditions in order to determine adequacy of reel design and shall be rewound after use.

4.6.9 Halon system test. The Halon system shall be tested to determine compliance with 3.11.2.17. Nozzle discharge flow rate and stream range shall be met. During operations, the control shall be checked for adequacy and ease of operation. The hose shall be run fully out under simulated firefighting conditions in order to determine adequacy of reel design and shall be rewound after use.

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4.6.10 Auxiliary equipment. All portable extinguishers, ladders, and tools shall be removed from their mounting brackets in order to determine their ready availability.

4.6.11 Drain system. The trucks agent drain system shall be demonstrated for proper operation.

4.7 First article vehicular performance tests.

4.7.1 Weight determination. The gross and curb weights of the truck shall be determined when loaded as specified in 3.6.

4.7.2 Determination of load distribution. The loads on each wheel and each axle shall be determined at curb and gross weight for compliance with 3.8.2.

4.7.3 Determination of turning clearance and turning characteristics. A marker shall be so placed that it will describe the largest circle in a turn at maximum cramping angle. The truck shall then be moved with steering cramped at maximum angle to the left (marker located on the right side) until a circle has been inscribed on the ground. The diameter of this inscribed circle represents the turning clearance to the left. This procedure shall be repeated with the marker on the left side of the truck to obtain the turning clearance to the right. The truck with the agent tanks 75 percent full shall then be subjected to sharp turns at various speeds in both directions and the reaction of the truck observed.

4.7.4 Determination of limits of vision. The limits of vision of the driver and equipment operator shall be determined by direct visions as specified in 3.9.18.2.1, seated in the truck, in their positions and using a large pad marked with concentric circle to indicate distance to right or left of normal. Blind spots within these fields of vision shall be noted.

4.7.5 Vehicle performance tests. The total gross weight of the complete fire truck shall not exceed the manufacturer's gvw rating when fully equipped as specified herein, including all specified firefighting equipment serviced with fuel, water, and lubricants; full water and foam tanks; a crew in the cab of four men weighing not less than 175 lb each or equivalent deadweight. The weight distribution shall be as specified herein. The truck, when tested as described below, shall comply with the performance requirement specified in 3.6. The truck shall be tested over prepared and selected test courses to determine the vehicular performance of the test item.

4.7.5.1 Maximum speed. The truck, with gross load, shall be operated on dry, level concrete roadway to determine maximum speed. A total of at least 10 test runs (five in opposite directions) shall be conducted. Maximum speed will be the average value obtained.

4.7.5.2 Acceleration. The time-speed relation of the truck with gross load shall be determined while accelerating from a standstill to its maximum speed on level concrete highway. A total of at least 10 test runs (five in opposite directions) shall be conducted. Acceleration will be the average value obtained.

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4.7.5.3 Grade speeds. The truck, with gross load, shall be driven up a grade of not less than 30 percent, and the maximum speed shall be determined. Engine oil and fuel pressure shall be observed and engine starting and idling ability shall be determined. The truck is to ascend a 10 percent grade at a maintained speed of 20 mph.

4.7.5.4 Side slope operation. The 20 percent side slope operational characteristics (both right and left) of the truck with gross load shall be determined upon courses prepared for testing 20 percent side slope operation. Firefighting equipment engagement and water discharge characteristics shall be determined during these tests. Any excessive leakage from agent tanks shall be noted. The vehicle shall be parked on a 30 percent grade and the engine shall be stopped, started, and idled for 2 minutes.

4.7.5.5 Drawbar pull and cooling. In an ambient air temperature not lower than 60°F, the truck with gross load shall be operated on dry pavement under road-load conditions until normal operating temperatures of the power train components have been reached. Utilizing a mobile dynamometer, the truck shall then be operated for 1 hour in low gear at 70 percent of maximum drawbar pull or until one of the component temperatures exceeds these critical values:

a. Engine coolant--95°F above ambient extrapolated to 125°F.
b. Engine oil--150°F above ambient extrapolated to 125°F.
c. Gear box oil--175°F above ambient extrapolated to 125°F.

Temperatures obtained shall be extrapolated on a one-for-one basis for 125°F.

4.7.5.6 Braking ability. The truck, with gross load, shall be brought to a stop with the service brakes when headed both up and down a dry, hard surfaced, 60 percent grade. After the truck has been stopped when headed both up and down on a 60 percent grade, the handbrake shall be applied and the service brakes released. The truck shall remain stationary. The truck shall be driven at a speed of 20 mph on level concrete, and the minimum distance required to bring the truck to a stop shall be measured. This shall be accomplished by appropriate methods which will eliminate the reaction time of the driver. The capability of the service brakes to fully recover within 3 minutes at the end of a run, down a 3-mile (average 10 percent) grade with transmission in high gear and road speed controlled by repeated application of brakes not to exceed 30 mph, shall be demonstrated. In lieu of the 3-mile, 10 percent grade, brake fade test described above, the capability of the service brake system to deliver 70 percent full braking capacity to the brakes when applied at a rate of three applications per minute with transmission in high gear and engine at maximum governed speed shall be demonstrated.

4.7.5.7 Obstacle crossing ability. The truck with gross load shall be tested over an 18-inch vertical wall obstacle course. Any interference between vehicle components or between the vehicle and obstacle profile shall be noted and corrected. To further check for frame, suspension, and drive train flexibility, the truck at gross load shall be jacked up to allow placement of blocks under the wheels equal in height to the cross section diameter of the tires. The truck shall be in a static position and the blocks shall be placed under opposite wheels in alternately contrary directions.

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4.7.5.8 Vehicle CI. To determine compliance to 3.6(i), the vehicle with gross load shall traverse terrain consisting of unflooded, fine-grained, dry-to-moist soils having CI rating in the critical layer no greater than 45° as specified in 6.5. Calculations will be acceptable to verify compliance.

4.7.5.9 Road and cross-country. The truck shall be subjected to two cycles of operation. One cycle with the agent tanks full and one cycle of operation with the agent tanks 50 percent full over the following roads and terrain at the speeds and distances specified:

Cycle	Average speed (mph)	Maximum speed (mph)	Distance (miles)
Hard surface highways	40	60	200
Gravel roads	25	35	100
Cross-country level	15	20	100
Cross-country hilly	15	20	100

4.8 First article environmental tests.

4.8.1 Environmental temperature test data. During the environmental operational tests, temperatures at least at the following points shall be recorded at intervals not greater than 15 minutes:

a. Ambient--four dispersed places.
b. Coolant inlet and outlet at booster heater.
c. Coolant inlet for all major components and compartments being heated.
d. Air at center and bottom of all compartments being heated.
e. Air in cab at air inlet, operator's foot level, center, and ceiling.
f. Windshield at center.
g. Agent in center of agent tanks.
h. Water at recirculating pump, and all nozzle shutoff valves.
i. Foam concentrate at pump outlet, and all inductors.

4.8.2 Cold test. The truck shall be serviced with the proper low temperature greases, oils, and winter coolants; filled with fuel and agents. The agent system shall be drained and the drain time recorded. The truck shall be instrumented in accordance with 4.8.1 and the system inspected for undrained pockets of agent. The truck shall be placed in a cold chamber using refrigerated air and the temperature within the chamber lowered and maintained at 0°±2°F for a period of 24 hours or until temperature stabilization occurs. The capability to start the engine at 0°F without use of the winterization system shall be demonstrated. The cold room temperature shall be lowered and maintained at -25°±2°F for a period of 24 hours or until temperature stabilization occurs. During the period, critical temperatures shall be recorded. At the completion of the cold soak period, the agent tanks shall be filled and the vehicle's heating system activated (both engine, power plant heater, and the truck's heater system described in paragraphs 3.9.18.5 and 3.9.18.5.1). At the end of 1 hour operation of the vehicle's heating system, the capability of the truck to start within 15 seconds at -25°F shall be demonstrated. Also at this time, the vehicle firefighting system shall be checked by momentarily operating all agent dispensing nozzles. Critical

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temperatures shall be recorded at this time. For a minimum period of 4 hours, at least two people shall occupy the cab to observe cab heating and window defrosting system operation, cab instrument operation, opening and closing cab windows and doors, and to record pertinent data and note any corrective action required for crew comfort during simulated standby operation for this period at -25°F. Also, during this test period a small amount of agent shall be drained from each drain in the agent system for evidence of any freezing conditions. At the end of the 4-hour period the agent system shall again be momentarily operated. There shall be no evidence of freeze damage to any of the systems or components during the test.

4.8.3 High temperature operation. The truck shall be filled with agent, instrumented in accordance with 4.8.1 (except for winterization coolant system temperatures), placed in a hot chamber, and subjected to an ambient temperature of 125°±2°F for a period of at least 24 hours or until temperature stabilization is reached. The truck engine and pump engine shall be started, and all component equipment including the air conditioning checked to determine that they will operate satisfactorily at the 125°F temperature. Agent dispensing equipment shall be tested by discharging the agent simultaneously at rated capacity through the various nozzles. At least one complete filling of the water tank shall be discharged. With the ambient temperature maintained at 125°F ±2°F, the fire pump shall be operated to discharge 500 gpm of water continuously for a 2-hour period to assure proper truck engine cooling capacity when the truck is used as a pumper.

4.8.4 Air conditioner test. After high temperature operation at 125°F, the temperature of the chamber shall be lowered to 115°F ±2°F and maintained for a period of at least 24 hours or until temperature stabilization is reached. The air conditioner shall be started. During this test, a crew of four shall occupy the truck or their metabolic heat load simulated. The cab temperature shall be lowered to 85°F and the time to reach this temperature recorded. The 85°F cab temperature shall be maintained for 2 hours. At the end of this test, all safety controls shall be checked for effectiveness. The air conditioner shall exhibit no malfunctioning, leaks, rough operation, vibration, or other irregularities as a result of this test.

4.8.5 Radio interference suppression test. To determine conformance to 3.16, the first article model shall be tested as specified in SAE J551 or the manufacturer shall furnish certified test data on an identically equipped vehicle that has been tested in accordance with SAE J551. Certified test data shall be accomplished by a description of all radio suppression devices used, adequate for visual determination that the first article model is identically equipped.

4.8.6 Noise level test. Noise level tests shall be conducted for compliance with 3.6.2 and 3.9.18.6.6.

4.8.7 Rain tests. Rain tests specifically for demonstrating that cab construction is rainproof shall be conducted. The cab shall be tested to a water spray simulating a rainfall of 3 inches per hour for a period of 1 hour to determine leakage.

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4.9 Production sample. Upon acceptance of the first production vehicle, it shall remain at the manufacturing facility as a production sample, and may be the last vehicle shipped on the contract. The contractor shall maintain the vehicle in a serviceable condition for the duration of the contract.

4.10 Production unit tests. Each production truck shall be subject to the following tests:

4.10.1 One hundred mile road test. The truck, with agent tanks 75 percent full, shall be operated for a distance of 50 miles during which 5 miles shall be in the 55 mph range; 20 miles at an average of 40 mph; 20 miles at a speed of 20-30 mph; and 5 miles in the 5-10 mph range. During the test, the following functions shall be performed:

a. Normal or average stops, 10 times.
b. Quick stops from a speed of at least 20 mph by instantaneous application of full service braking power, five times.
c. Sharp right turns and sharp left turns, five times each.
d. Operation through all forward and reverse gears including the transfer case if applicable.
e. Differential locking device and all wheel drive shall be engaged for at least 5 miles of operation.

4.10.2 Turret systems and water flow rate tests. Water flow rate to be conducted in accordance with 4.6.4.

4.10.3 Turret. The roof and bumper turret system controls shall be tested for performance and ease of operation as set forth in 4.6.6 and 4.6.7.

4.10.4 Handline system. The handline system controls shall be tested for performance and ease of operation. The handline shall be run fully out under simulated emergency conditions to determine adequacy of reel design then rewound after use. During this test, the handline nozzle attachments shall be checked for water use only for flow rates as specified in 3.11.2.16.

4.10.5 Halon system. The Halon system controls shall be checked for performance and ease of operation. The hose shall be run fully out to determine satisfactory reel design and assembly.

4.10.6. Rejection and retest. When an item selected from production run fails to meet the specification, no items still on hand or later produced shall be accepted until the extent and cause of failure have been determined and appropriately corrected. The contractor shall explain to the government representative the cause of failure and the action taken to preclude recurrence. After correction, all tests influenced by the failed components shall be repeated.

4.11 Packaging inspection. The vehicle shall be inspected to verify conformance to the requirements of section 5.

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5. PACKAGING

* 5.1 Vehicle processing. The equipment shall be preserved and packaged in accordance with the contractors standard practice. When specified (see 6.2), the equipment shall be preserved and packaged in accordance with the mobile requirements of MIL-STD-281 with the level of preservaton and packaging as specified (see 6.2).

6. NOTES

6.1 Intended use. The firefighting truck is intended to be used for hydrocarbon fire suppression and personnel rescue work following aircraft accidents.

* 6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Specifications and standards (see 2.1.1).
c. When a first article inspection is required (see 3.2 and 4.2.1).
d. When preservation and packaging shall be in accordance with MIL-STD-281; and the level of preservation and packaging required (see 5.1).

6.3 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL) and invokes the provisions of paragraph 52.227-7031 of the Federal Acquisition Regulations (FAR), the data requirements will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL (DD Form 1423) incorporated into the contract. When the provisions of FAR 52.227-7031 are not invoked, the data shall be delivered in accordance with the contract requirements.

* 6.4 First article. When a first article inspection is required (see 3.2 and 6.2), the item will be tested and should be a first production item or it may be a standard production item from the contractor's current inventory as specified in 4.2.1. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.5 Definitions.

6.5.1 Cross-country terrain. Cross-country terrain consists of open fields, broken ground, loose sand or gravel road, and uneven terrain in general.

6.5.2 Vehicle CI. Personnel from the Waterways Experiment Station, U.S. Army Corps of Engineers, Vicksburg, Mississippi, have empirically developed a relationship between soil types and vehicle mobility with the aid of the cone penetrometer. The procedure for cone penetrometer measurements is reflected in the Department of the Army Technical Bulletin, TB ENG 37, dated 10 July 1959; and a means of predetermining vehicles mobility characteristics by computation is contained in U.S. Army Material Command Technical Memorandum No. 3-240 (18th supplement), date March 1968. In addition, vehicle CI

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requirements have been developed for a large number of representative U.S. Army vehicles. The following definitions pertain to this system:

a. Cone penetrometer. A device used in evaluating trafficability of soils. It consists of a 36-inch long metal shaft with a 30° right circular cone of 0.5 square inch base area mounted on one end of the shaft.
b. Cone index. A dimensionless value read from the dial gage of the cone penetrometer. The CI is considered an index of shear resistance of the soil.
c. Vehicle CI. The minimum CI of a soil in the critical layer no greater than 45° which will permit a specific vehicle to make 50 passses without immobilization. It can be determined by first computing a mobility index from an empirical formula given in Appendix A of Technical Memorandum No. 3-240 and referring to a conversion table given in the same appendix.
d. Sheargraph. An instrument that measures surface traction properties of soils at normal pressures equal to the average tire contact pressures on a firm surface.

6.6 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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User activity:

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-T-21737F	TRUCK, FIREFIGHTING: AIRCRAFT AND PERSONNEL RESCU... WATER AND FOAM DISPENSING, 1,000-GALLON WATER CAPACITY

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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